



SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(Autonomous)

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Dr.V. Nirupama





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National Conference on

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ETMS -2025 in Hybrid Mode

Date : 29-04-2025 **Time** : 10:00 AM

Venue : Smt. Lakshmi Devi Memorial Auditorium

Organised by
Department of Humanities & Sciences



Physical Properties of Rapid Thermal Annealed Vacuum Evaporated Zinc Phthalocyanine (ZnPc) Thin films

T. Venkata Raju¹, A. Ashok Kumar^{1,*}, S. Kaleemulla², V. Rajagopal Reddy³

¹Department of Physics, YSR Engineering College of Yogi Vemana University, Proddatur, YSR Kadapa (Dt),

Andhra Pradesh, India.

²Thin Films Laboratory, Centre for Functional Materials, Vellore Institute of Technology, Vellore, Tamil Nadu, India.

³Department of Physics, Sri Venkateswara University, Tirupati, Andhra Pradesh, India.

*Corresponding Author: drashok.yvuce@gmail.com

ABSTRACT

Influence of annealing on the physical properties such as optical, structural and morphological properties of thermally evaporated Zinc Phthalocyanine (ZnPc) thin films were analysed. Optical properties of as-deposited ZnPc thin films and thin films after rapid thermal annealing at 100 °C, 200 °C, 300 °C and 400 °C reveals a notable Q-band (Q_1 at 638 nm and Q_2 at 680 nm) splitting due to electronic transition from $\pi \rightarrow \pi^*$ orbitals. Structural properties of ZnPc with and without annealing were discussed using Raman spectroscopy. A clear evidence of significant molecular vibrational bands at 1508 cm^{-1} , 1430 cm^{-1} and 1338 cm^{-1} observed in as-deposited ZnPc thin films might be related to pyrrole stretching bands. Considerable shifts of these bands are observed towards to lower wavenumber at 300 °C annealing temperature signifies a change of ZnPc thin film from α -phase to β -phase. A similar type of observation is also evidenced in optical properties of the films indicating a variation in the absorption intensity in Q_1 band than comparing to Q_2 band. Morphological properties ZnPc films with and without annealing is also studied using FESEM measurements (both surface and cross-section). A significant change in the morphology of ZnPc thin films annealed at 300 °C and 400 °C is observed with nanorods type structures. Cross-sectional view of ZnPc thin films on the n-Ge substrate with annealing clearly invades no diffusion ZnPc interlayer into substrate and also shows clear boundary between ZnPc and Ge interface. This evidences a highly stable ZnPc-Ge interface without any interdiffusion occurs at the boundary with rapid thermal annealing. The transformation of ZnPc thin films from α -phase to β -phase after 300 °C shows a variation in the structure and morphology of the thin films which in turn affects the electrical and optoelectronic properties of ZnPc/n-Ge heterojunction.

Keywords: Zinc Phthalocyanine (ZnPc), Rapid Thermal Annealing, Surface morphology, Optical properties, and Raman measurements.